Zachary P. Kilpatrick

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Professor of Applied Mathematics, University of Colorado Boulder

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Education

2007 – 2010 **University of Utah**: PhD in Mathematics

2005 – 2007 **University of Utah**: M.S. in Mathematics

2001 – 2005 **Rice University**: B.A. in Computational and Applied Mathematics; B.A. in History

Academic Appointments

2026 – **University of Colorado Boulder**, Professor, Applied Mathematics

2023 – 2026 **University of Colorado Boulder**, Graduate Program Chair, Applied Mathematics

2020 – 2026 **University of Colorado Boulder**, Associate Professor, Applied Mathematics

2016 – 2020 **University of Colorado Boulder**, Assistant Professor, Applied Mathematics

2012 – 2016 **University of Houston**, Assistant Professor, Mathematics

2010 – 2012 **University of Pittsburgh**, NSF Mathematical Sciences Postdoctoral Research Fellow

Affiliate appointments, University of Colorado Boulder: Computer Science (2023–), Institute of Cognitive Science (2018–), BioFrontiers Institute (2016–), Center for Neuroscience (2016–). University of Colorado School of Medicine: Physiology & Biophysics (2016–). National Institute for Theory and Mathematics in Biology, NITMB (2026–).

Current Research Grants

amount to Kilpatrick in **bold**

2026 – 2029 NSF CISE Future Computing Research (PI: **$528,146**; IIS-2616531)
RI: Principles of robust and flexible computation under resource constraints

2025 – 2028 NSF Emerging Mathematics in Biology (PI with O. Peleg: **$155,000**; DMS-2527338)
Collective and multiscale dynamics of resource transfer in social insects

Completed Research Grants

amount to Kilpatrick in **bold**

2022 – 2025 NSF CRCNS Program (co-PI with J. Gold, L. Ding, & K. Josić: **$242,421**; NSF-DMS-2207700)
CRCNS: Adaptive decision rules in dynamic environments

2020 – 2024 NIH BRAIN Initiative, NIMH/NIBIB (PI: **$772,372**; R01-EB029847)
Connecting neural circuit architecture and experience-driven probabilistic computations

2019 – 2022 NSF DMS: Mathematical Biology (PI: **$249,999**; DMS-1853630)
Spatiotemporal neural dynamics of visual decisions

2017 – 2021 NSF/NIH CRCNS Program (co-PI with J. Gold & K. Josić: **$532,732**; R01-MH115557)
CRCNS: Decision making in changing environments

2016 – 2019 NSF DMS: Mathematical Biology (PI: **$234,000**; DMS-1615737)
Robust spatiotemporal dynamics in multi-layer neuronal networks

2015 – 2019 NSF DMS: Mathematical Biology (co-PI with K. Josić: **$164,722**; DMS-1517629)
The ever-changing network: How changes in architecture shape neural computations

2013 – 2017 NSF DMS: Mathematical Biology (PI: **$184,937**; DMS-1311755)
Architecture for robust spatiotemporal dynamics in neuronal networks

2010 – 2012 NSF DMS: Postdoctoral Research Fellowship (PI: **$135,000**)

Conference Grants, Internal Grants, Awards, & Honors

awarded amount in **bold**

2025 – 2026	CU Boulder Faculty Leadership Institute
2024	CU Boulder Graduate School Outstanding Faculty Mentor Award: **$700**
2016 – 2017	*International Conference on Mathematical Neuroscience* NSF DMS Conference Proposal (PI with J. Gjorgjieva & R. Rosenbaum: **$20,000**) Burroughs Wellcome Fund Conference Proposal (co-PI with J. Gjorgjieva: **$5,000**) SIAM Conference Proposal (PI with J. Gjorgjieva & R. Rosenbaum: **$5,000**) CU Boulder Faculty Conference Award: **$3,000**
2013 – 2014	University of Houston, GEAR (co-PI with K. Josić: **$30,000**) *Forecasting in biological networks: How organisms see the future*
2013	University of Houston, New Faculty Research Grant (sole PI: **$6,000**) *Robust neural field models for decision making with multiple alternatives*

Graduate Students Supervised

5 current & 7 past PhD; 2 current & 3 past masters

Nirab Hossain	PhD, exp 2030	CU Boulder
Pedro Albuquerque Lemos*	PhD, exp 2029	CU Boulder, CSCI
Noah Palmer	PhD, exp 2029	CU Boulder
William Magrogan*	PhD, exp 2027	CU Boulder
Noah Parks	PhD, exp 2027	CU Boulder
Sage Shaw	PhD 2025	Postdoc, U Minnesota
Heather Cihak	PhD 2024	Visiting Assistant Professor, U Minnesota
Nicholas Barendregt	PhD 2023	Data Scientist, Boulder Imaging
Subekshya Bidari	PhD 2022	Data Scientist, CDC
Kate Nguyen*	PhD 2020	Data Science Consultant
Adrian Radillo*	PhD 2018	Data Scientist, Chubb
Daniel Poll	PhD 2017	Assistant Professor, College of Charleston
Zara Chandra	M.S., exp 2027	CU Boulder
Matthew Younce	M.S., exp 2027	CU Boulder
Emily Webb	M.S. 2021	Systems Engineer, JHU APL
Timothy Thorn	M.S. 2020	Manager, Centene Corporation
Nikhil Krishnan	M.S. 2019	PhD Student, Princeton ORFE

*Coadvised: Lemos (O. Peleg), Magrogan (N. Rodriguez), Nguyen & Radillo (K. Josić).
Fellowships: Palmer & Lemos (NIH BioDataSci T32); Cihak (NSF GRF Honorable Mention); Bidari (AAUW, DAAD); Nguyen (NSF GRF).

Postdoctoral Fellows Supervised

Ryan Thorpe	2024–	CU Anschutz/Boulder
Tahra Eissa	2018–2025	Asst Prof Physiology, CU Anschutz
Alan Veliz-Cuba	2013–2015	Assoc Prof Math, U Dayton

K99/R00 BRAIN Initiative Career Transition Award ($1,010,710): Eissa. Coadvised with K. Josić: Veliz-Cuba.

Manuscripts under Review [link]

undergrad*; grad student‡; postdoc†; co-first$^{\oplus}$; co-last$^{\bullet}$

1. J. Bosulu, V. Balasubramanian, Z.P. Kilpatrick, K. Josić, & J.I. Gold, *An information compass for probabilistic reasoning in the brain*, **eLife** (2026) in rev.
   bioRxiv: `https://www.biorxiv.org/content/10.64898/2026.10.07.757408v1`
2. Z.P. Kilpatrick, *Social discounting enables fast and reliable collective escape*, **PNAS** (2026) in rev.
   arXiv: `https://arxiv.org/abs/2608.14965`
3. Z.P. Kilpatrick & A. El Hady, *Resource depletion accelerates rate learning but not composition learning in patch foraging*, **PLoS Comput. Biol.** (2026) in rev.
   arXiv: `https://arxiv.org/abs/2607.29476`

Refereed Journal Publications

1. C. Jarne, R. Yoon, T.L. Eissa†, Z.P. Kilpatrick, & K. Josić, *Task-parametrized dynamics: Representation of time and decisions in recurrent neural networks*, **J. Comput. Neurosci.** (2026) in press.
2. N. Parks‡ & Z.P. Kilpatrick, *Stroboscopic motion reversals in delay-coupled ring neural fields*, **SIAM J. Appl. Dyn. Syst.** (2026) in press.
3. M. Yerlanov‡, Z.P. Kilpatrick, & N. Rodriguez, *Modularity, asymmetry, and polarization shape consensus speed in the voter model*, **Phys. Rev. E** 114 (2026) 044313.
4. N. Palmer‡, H.L. Cihak‡, D. Avitabile, & Z.P. Kilpatrick, *Astrocytic resource diffusion stabilizes persistent activity in neural fields*, **SIAM J. Life Sci.** 1 (2026) pp. 485-508.
5. E.D. Prévost, L.A. Ward, D. Alas, G.A. Aimale, S.K. Ikenberry, K. Fox, M. Deming, J. Pelletier, A. Ly, J. Ball, Z.P. Kilpatrick, K. Price, A.M. Polter, F. Simões de Souza, S. Fulgham, Y.S. Kim, L.E. Fenno, C. Ramakrishnan, K. Deisseroth, & D.H. Root, *Distinct properties of glutamate-dopamine, nonglutamate-dopamine, and glutamate-only ventral tegmental area neurons*, **Sci. Signal. 19** (951) (2026), eaee2028.
6. H. Kim‡, Z.P. Kilpatrick, & K. Josić, *Daring few, patient many: Division of labor in decentralized foraging collectives*, **PNAS** 123 (2026) e2605309123.
7. I. Kalburge$^{\oplus}$, A. Dallstream$^{\oplus}$, K. Josić, Z.P. Kilpatrick, L. Ding, & J.I. Gold, *Human decision-makers terminate evidence accumulation using flexible decision rules*, **eLife** 11 (2026) e111082.
8. N.W. Barendregt, J.I. Gold, K. Josić, & Z.P. Kilpatrick, *Information-seeking decision strategies mitigate risk in dynamic, uncertain environments*, **J. R. Soc. Interface** 23 (2026) 20251308.
9. S.B. Shaw‡, Z.P. Kilpatrick, & D. Avitabile, *Radial basis function techniques for neural field models on surfaces*, **SIAM J. Sci. Comp.** 48 (2026) pp. A668-A674.
10. H.L. Cihak‡ & Z.P. Kilpatrick, *Robustly encoding certainty in a metastable neural circuit model*, **Phys. Rev. E** 110 (2024) 034404.
11. H.L. Cihak‡ & Z.P. Kilpatrick, *Multiscale motion and deformation of bumps in stochastic neural fields with dynamic connectivity*, **Multiscale Model. Simul.** 22 (2024) pp. 178-203.
12. T.L. Eissa† & Z.P. Kilpatrick, *Learning efficient representations of heterogeneity in attractors for working memory*, **PLoS Comput. Biol.** 19 (2023) e1011622.
13. A. Ly, A. Barker, E.D. Prevost, D.J. McGovern, Z.P. Kilpatrick, & D.H. Root, *Bed Nucleus of the Stria Terminalis GABA neurons are necessary for changes in foraging behavior following an innate threat*, **Eur. J. Neurosci.** 58 (2023) pp. 3630-3649.

14. M. Stickler, W. Ott, Z.P. Kilpatrick, K. Josić, & B. Karamched, *Impact of correlated information on pioneering decisions*, **Phys. Rev. Research** 5 (2023) 033020.
15. N.W. Barendregt‡, E.G. Webb*, & Z.P. Kilpatrick, *Adaptive Bayesian inference of Markov transition rates*, **Proc. R Soc. A** 479 (2023) 20220453.
16. J.I. Gilmer, M.A. Farries, Z.P. Kilpatrick, I. Delis, & A.L. Person, *An emergent temporal basis set robustly supports cerebellar time-series learning*, **J. Neurophysiol.** 129 (2023) pp. 159-176.
17. N.W. Barendregt‡, J.I. Gold, K. Josić, & Z.P. Kilpatrick, *Normative decision rules in changing environments*, **eLife** 11 (2022) e79824.
18. H.L. Cihak‡, T.L. Eissa†, & Z.P. Kilpatrick, *Distinct excitatory and inhibitory bump wandering in a stochastic neural field*, **SIAM J. Appl. Dyn. Syst.** 21 (2022) pp. 2579-2609.
19. T.L. Eissa†, J.I. Gold, K. Josić, & Z.P. Kilpatrick, *Suboptimal human inference inverts the bias-variance trade-off for decisions with asymmetric evidence*, **PLoS Comput. Biol.** 18 (2022) e1010323.
20. S. Bidari‡, A. El Hady, J.D. Davidson, & Z.P. Kilpatrick, *Stochastic dynamics of social patch foraging decisions*, **Phys. Rev. Research** 4 (2022) 033128.
21. K Schapiro, K. Josić, Z.P. Kilpatrick, & J.I. Gold, *Strategy-dependent effects of working-memory limitations on human perceptual decision-making*, **eLife** 11 (2022) e73610.
22. S. Bidari‡ & Z.P. Kilpatrick, *Hive geometry shapes the recruitment rate of honeybee colonies*, **J. Math. Biol.** 83 (2021) 20.
23. Z.P. Kilpatrick, J.D. Davidson, & A. El Hady, *Uncertainty drives strategy deviations of patch leaving decisions in foraging*, **J. R. Soc. Interface** 18 (2021) 20210337.
24. B. Karamched†, M. Stickler‡, W. Ott, B. Lindner, Z.P. Kilpatrick, & K. Josić, *Heterogeneity improves speed and accuracy in social networks*, **Phys. Rev. Lett.** 125 (2020) 218302. [**Highlight**] and [**Editors' Suggestion**]
25. B. Karamched†,⊕, S. Stolarczyk‡,⊕, Z.P. Kilpatrick, & K. Josić, *Bayesian evidence accumulation on social networks*, **SIAM J. Appl. Dyn. Syst.** 19 (2020) pp. 1884-1919.
26. Y. Wang, Z.P. Kilpatrick, & K. Josić, *A hierarchical model of perceptual multistability involving interocular grouping*, **J. Comput. Neurosci.** 48 (2020) pp. 177-192.
27. S. Bidari‡, O. Peleg, & Z.P. Kilpatrick, *Social inhibition maintains adaptivity and consensus of foraging honey bees in dynamic environments*, **R. Soc. Open Sci.** 6 (2019) 191681.
28. N.W. Barendregt‡, K. Josić, & Z.P. Kilpatrick, *Analyzing dynamic decision-making models using Chapman-Kolmogorov equations*, **J. Comput. Neurosci.** 47 (2019) pp. 205-222.
29. A.E. Radillo‡,⊕, A. Veliz-Cuba⊕, K. Josić, & Z.P. Kilpatrick, *Performance of normative and approximate evidence accumulation on the dynamic clicks task*, **Neurons, Behavior, Data Analysis, & Theory** (2019) 10226.
30. Z.P. Kilpatrick, W.R. Holmes, T.L. Eissa†, & K. Josić, *Optimal models of decision-making in dynamic environments*, **Curr. Opin. Neurobiol.** 58 (2019) pp. 54-60.
31. K.P. Nguyen‡, K. Josić, & Z.P. Kilpatrick, *Optimizing sequential decisions in the drift-diffusion model*, **J. Math. Psychol.** 88 (2019) pp. 32-47.
32. N. Krishnan* & Z.P. Kilpatrick, *Optimizing a jump-diffusion model of a starving forager*, **Phys. Rev. E** 98 (2018) 052406.
33. G. Faye & Z.P. Kilpatrick, *Threshold of front propagation in neural fields: An interface dynamics approach*, **SIAM J. Appl. Math.** 78 (2018), pp. 2575-2596.
34. Z.P. Kilpatrick, *Synaptic mechanisms of interference in working memory*, **Sci. Rep.** 8 (2018) 7879.

35. N. Krishnan*, D.B. Poll‡, & Z.P. Kilpatrick, *Synaptic efficacy shapes resource limitations in working memory*, **J. Comput. Neurosci.** 44 (2018), pp. 273-295.

36. Z.P. Kilpatrick & D.B. Poll‡, *Neural field model of memory-guided search*, **Phys. Rev. E** 96 (2017), 062411.

37. D.B. Poll‡ & Z.P. Kilpatrick, *Velocity integration in a multilayer neural field model of spatial working memory*, **SIAM J. Appl. Dyn. Syst.** 16 (2017), pp. 1197-1234.

38. A.E. Radillo‡, A. Veliz-Cuba, K. Josić, & Z.P. Kilpatrick, *Evidence accumulation and change rate inference in dynamic environments*, **Neural Comput.** 29 (2017), pp. 1561-1610.

39. A. Jacot-Guillarmod, Y. Wang, C. Pedroza, H. Öğmen, Z.P. Kilpatrick, & K. Josić, *Extending Levelt's Propositions to perceptual multistability involving interocular grouping*, **Vision Res.** 133 (2017), pp. 37-46.

40. Z.P. Kilpatrick, *Ghosts of bump attractors in stochastic neural fields: Bottlenecks and extinction*, **Discrete Contin. Dynam. Syst. Ser. B** 21 (2016), pp. 2211-2231.

41. Z.T. McCleney* & Z.P. Kilpatrick, *Entrainment in up and down states of neural populations: non-smooth and stochastic models*, **J. Math. Biol.** 73 (2016), pp. 1131-1160.

42. D.B. Poll‡ & Z.P. Kilpatrick, *Persistent search in confined domains: a velocity-jump process model*, **J. Stat. Mech.** (2016), 053201.

43. D.B. Poll‡, K. Nguyen*, & Z.P. Kilpatrick, *Sensory feedback in a bump attractor model of path integration*, **J. Comput. Neurosci.** 40 (2016), pp. 137-155.

44. A. Veliz-Cuba†, Z.P. Kilpatrick, & K. Josić, *Stochastic models of evidence accumulation in changing environments*, **SIAM Rev.** 58 (2016), pp. 264-289.

45. A. Veliz-Cuba†, H.Z. Shouval, K. Josić, & Z.P. Kilpatrick, *Networks that learn the precise timing of event sequences*, **J. Comput. Neurosci.** 39 (2015), pp. 235-254.

46. D.B. Poll‡ & Z.P. Kilpatrick, *Stochastic motion of bumps in planar neural fields*, **SIAM J. Appl. Math.** 75 (2015) pp. 1553-1577.

47. Z.P. Kilpatrick, *Stochastic synchronization of neural activity waves*, **Phys. Rev. E** 91 (2015), 040701(R).

48. P.C. Bressloff & Z.P. Kilpatrick, *Nonlinear Langevin equations for wandering patterns in stochastic neural fields*, **SIAM J. Appl. Dyn. Syst.** 14 (2015), pp. 305-334.

49. Z.P. Kilpatrick, *Delay stabilizes stochastic motion of bumps in layered neural fields*, **Physica D** 295 (2015), pp. 30-45.

50. Z.P. Kilpatrick & G. Faye, *Pulse bifurcations in stochastic neural fields*, **SIAM J. Appl. Dyn. Syst.** 13 (2014), pp. 830-860.

51. J.K. Kim†, Z.P. Kilpatrick, M.R. Bennett, & K. Josić, *Molecular mechanisms that regulate the coupled period of the mammalian circadian clock*, **Biophys. J** 106 (2014), pp. 2071-2081.

52. Z.P. Kilpatrick, *Coupling layers regularizes wave propagation in stochastic neural fields*, **Phys. Rev. E** 89 (2014), 022706.

53. S. Carroll*, K. Josić, & Z.P. Kilpatrick, *Encoding certainty in bump attractors*, **J. Comput. Neurosci.** 37 (2014), pp. 29-48.

54. Z.P. Kilpatrick, B. Ermentrout, & B. Doiron, *Optimizing working memory with heterogeneity of recurrent cortical excitation*, **J. Neurosci.** 33 (2013), pp. 18999-19011.

55. Z.P. Kilpatrick, *Interareal coupling reduces encoding variability in multi-area models of spatial working memory*, **Front. Comput. Neurosci.** 7 (2013), 82.

56. Z.P. Kilpatrick & B. Ermentrout, *Wandering bumps in stochastic neural fields*, **SIAM J. Appl. Dyn. Syst.** 12 (2013), pp. 61-94.

57. Z.P. Kilpatrick, *Short term synaptic depression improves information transfer in perceptual multistability*, **Front. Comput. Neurosci.** 7 (2013), 85.

58. S.M. Jayasuriya* & Z.P. Kilpatrick, *Effects of time-dependent stimuli on a competitive neural network model of perceptual rivalry*, **Bull. Math. Biol.** 6 (2012), pp. 1396-1426.

59. Z.P. Kilpatrick & B. Ermentrout, *Response of traveling waves to transient inputs in neural fields*, **Phys. Rev. E** 85 (2012), 021910.

60. Z.P. Kilpatrick & G.B. Ermentrout, *Hallucinogen persisting perception disorder in neuronal networks with adaptation*, **J. Comput. Neurosci.** 32 (2012), pp. 25-53.

61. Z.P. Kilpatrick & G.B. Ermentrout, *Sparse gamma rhythms arising through clustering in adapting neuronal networks*, **PLoS Comput. Biol.** 7 (2011), e1002281.

62. P.C. Bressloff & Z.P. Kilpatrick, *Two–dimensional bumps in piecewise smooth neural fields with synaptic depression*, **SIAM J. Appl. Math.** 71 (2011), pp. 379-408.

63. Z.P. Kilpatrick & P.C. Bressloff, *Binocular rivalry in a competitive neural network model with synaptic depression*, **SIAM J. Appl. Dyn. Syst.** 9 (2010), pp. 1303-1347.

64. Z.P. Kilpatrick & P.C. Bressloff, *Stability of bumps in piecewise smooth neural networks with nonlinear adaptation*, **Physica D** 239 (2010), pp. 1048-1060.

65. Z.P. Kilpatrick & P.C. Bressloff, *Spatially structured oscillations in a two-dimensional excitatory neuronal network with synaptic depression*, **J. Comput. Neurosci.** 28 (2010), pp. 193-209.

66. Z.P. Kilpatrick & P.C. Bressloff, *Effects of synaptic depression and adaptation on spatiotemporal dynamics of an excitatory neuronal network*, **Physica D** 239 (2010), pp. 547-560.

67. P.C. Bressloff & Z.P. Kilpatrick, *Nonlocal Ginzburg-Landau equation for cortical pattern formation*, **Phys. Rev. E** 78 (2008), 041916.

68. Z.P. Kilpatrick, S.E. Folias, & P.C. Bressloff, *Traveling pulses and wave propagation failure in inhomogeneous neural media*, **SIAM J. Appl. Dyn. Syst.** 7 (2008), pp. 161-185.

Editorials, Book Chapters, and Book Reviews (All Refereed)

B1. M. Aldana, R. Barak Ventura, H.Z. Brooks, P.S. Chodrow, F. Georgiou, J. Johnson, K. Josić, Z.P. Kilpatrick, K. Landgren, A. Nugent, M. Porfiri, N. Rodriguez, P. Suárez-Serrato, D. White, A. Wiedemann, & S. Zhang, *Modeling Social Systems: Transparency, Reproducibility, and Responsibility*, **SIAM News** (2026). **arXiv:** `https://doi.org/10.48550/arXiv.2508.18542`

B2. Z.P. Kilpatrick, *Featured Book Review: Neurodynamics: An Applied Mathematics Perspective (Kyle Wedgwood and Stephen Coombes)*, **SIAM Rev.** 66 (2024) pp. 391-394.

B3. Z.P. Kilpatrick, J Gjorgjieva, & R. Rosenbaum, *Special Issue from the 2017 International Conference on Mathematical Neuroscience*, **J. Math. Neurosci.** 9 (2019) 1.

B4. Z.P. Kilpatrick, *Featured Book Review: Methods and Models in Mathematical Biology (Johannes Muller and Christina Kuttler)*, **SIAM Rev.** 59 (2017) pp. 211-214.

B5. Z.P. Kilpatrick, *Wilson-Cowan model*, **Encyclopedia of Computational Neuroscience** (2014), Ed. D. Jaeger and R. Jung, Springer Verlag.

B6. G.B. Ermentrout, S.E. Folias, & Z.P. Kilpatrick, *Spatiotemporal pattern formation in neural fields with linear adaptation*, **Neural Field Theory** (2014), Ed. S. Coombes, P. beim Graben, R. Potthast and J.J. Wright, Springer Verlag.

Press

- *Researchers use honeybees to understand how organizations work* (9News, 8/2026)
- *Can leaderless societies get stuff done? Ask the honeybees* (CU Boulder Today, 7/2026)
- *Math reveals how honeybee hives balance the daring few, patient many strategy* (Phys.org, 7/2026)
- *When uncertainty spikes, chasing rewards backfires* (Phys.org, 5/2026)
- *Speed of decision-making reflects our biases* (New Scientist, 7/2024)
- *The data behind decision-making for voters* (Denver 7 News, 6/2022)
- *Diverse groups make better decisions* (Physics Today, 12/2020)
- *How laggards help decision-making* (APS Physics, 11/2020)
- *Roughly 240,000 Colorado voters changed their party affiliations* (Denver 7 News, 10/2020)
- *Election Day math* (CU Boulder Today, 10/2020)
- *Peer Pressure: How our social networks can change our choices* (APS Physics Buzz, 7/2020)
- *How people make Super Tuesday or other tough choices* (CU Boulder Today, 3/2020)

Curriculum Development

- **Brains & Machines: Modeling Intelligence with Neural Networks** (2025–, in development), *CU Boulder/Coursera.* Three-course specialization in computational neuroscience and biologically inspired AI.
- **Computational Neuroscience (APPM 4370/5370)** (2020–), *CU Boulder.* Designed cross-listed undergraduate/graduate course in Python-based neural modeling.
- **Tech Frontiers Executive Education** (2021–), *CU Boulder College of Engineering.* Developed modules on data science and machine learning.s

Teaching Experience

University of Colorado Boulder	**Term(s)**
APPM 4370/5370: Computational Neuroscience	F26, F24, F23, F21, S20
APPM 4560/5560: Markov Processes	F26, F25
APPM 4390/5390: Mathematical Biology	S26
APPM 7400: Graduate Research Seminar	S26
APPM 7400: Introduction to Research Seminar	S26, S25, S24
APPM 5470: Partial Differential & Integral Equations	F24, F20, F19, F18
APPM 8400: Mathematical Biology Seminar	S27, F26, F23, F19, S17
APPM 3010: Intro to Nonlinear Dynamics & Chaos	F21
APPM 5480: Approximation Methods	S21
APPM 2360: Differential Equations with Linear Algebra	S19
APPM 3570: Applied Probability	S18, S17
APPM 4350: Fourier Series & Boundary Value Problems	F16

Prior to CU Boulder	**Term(s)**
University of Houston: Mathematical Biology, Advanced Linear Algebra, Honors Engineering Mathematics	2012–2016
University of Pittsburgh: Analytic Geometry & Calculus I and II	2010–2011
University of Utah: Calculus for Biologists I and II	2007–2008

Invited Conference Plenary Talks

1. **Data-driven modeling, simulation, and inference for neurobiological problems** (NITMB), Chicago, Illinois, 9/2026
2. **Extreme events in biological functions** (NITMB), Chicago, Illinois, 6/2026
3. **Collective social phenomena: Dynamics and data**, Oaxaca, Mexico, 6/2025
4. **Topics on neuroscience, collective migration, and parameter estimation** (Oxford Mathematical Institute), Oxford, United Kingdom, 7/2023
5. **The dynamics of social interactions** (Aspen Center for Physics), Aspen, Colorado, 4/2022
6. **CRCNS Principal Investigators Meeting**, Austin, Texas, 9/2019
7. **Neuroethology of movement and motor control** (BIRS), Banff, Alberta, Canada, 5/2019
8. **International Neural Coding Workshop**, Torino, Italy, 9/2018
9. **International Conference on Mathematical Neuroscience**, Juan-les-Pins, France, 6/2018
10. **Winter School on Stochastic Models in Neuroscience**, Toulouse, France, 12/2017
11. **Undergraduate Capstone Conference** (MBI), Columbus, Ohio, 8/2017
12. **Brain dynamics and statistics: Simulation and data** (BIRS), Banff, Alberta, Canada, 2/2017
13. **Bernstein Sparks Workshop on Recurrent Network Theory**, Göttingen, Germany, 5/2016
14. **Connecting network architecture and computation** (BIRS), Banff, Alberta, Canada, 12/2015
15. **University of Texas Conference on Learning and Memory**, Austin, Texas, 4/2015
16. **Conference on Nonlinear Dynamics and Stochastic Methods**, Pittsburgh, Pennsylvania, 3/2014
17. **Gulf Coast Consortium Conference on Theoretical and Computational Neuroscience**, Houston, Texas, 1/2014
18. **Stochastic modeling of biological processes** (IMA), Minneapolis, Minnesota, 5/2013
19. **Conference on Progress in Neural Field Theory**, Reading, United Kingdom, 4/2012
20. **Spatio-temporal evolution equations and neural fields** (CIRM), Luminy, France, 10/2011

Departmental Colloquia and Seminar Talks

1. **Bernstein Center for Computational Neuroscience Berlin**, Berlin, Germany, 7/2026
2. **Brunel University of London, Equity, Diversity, & Inclusion Research in Action Seminar Series**, virtual, 5/2026
3. **Florida State University, Mathematics Colloquium**, Tallahassee, Florida, 3/2026
4. **University of Houston, Networks Seminar**, Houston, Texas, 2/2026
5. **University of Colorado Boulder, Neuroscience Seminar Series**, Boulder, Colorado, 10/2025
6. **University of Colorado Boulder, Cognitive Lunch Seminar**, Boulder, Colorado, 9/2025
7. **Max Planck Institute for Animal Behavior, Integrative Biophysics Seminar**, virtual, 4/2025
8. **University of Colorado, School of Medicine, Biomedical Informatics Seminar**, Aurora, Colorado, 11/2024
9. **University of Colorado School of Medicine, Physiology & Biophysics Seminar**, Aurora, Colorado, 12/2023

10. **Boston University, Dynamics Seminar**, Boston, Massachusetts, 10/2023
11. **Indiana University Purdue University Indianapolis, Math Biology REU Seminar**, virtual, 6/2023
12. **University of Pennsylvania, Center for Mathematical Biology**, Philadelphia PA, 4/2023
13. **Dartmouth College, Applied Mathematics Seminar**, Hanover NH, 9/2022
14. **Indiana University Purdue University Indianapolis, Math Biology REU Seminar**, virtual, 6/2022
15. **New Jersey Institute of Technology, Mathematics Colloquium**, Newark, New Jersey, 4/2022
16. **University of Vermont, Complex Systems & Data Science Seminar**, Burlington, Vermont, 4/2022
17. **Colorado School of Mines, Applied Mathematics Colloquium**, Golden, Colorado, 12/2021
18. **University of Exeter, Dynamics Seminar**, virtual, 9/2021
19. **University of Houston, Political Science Seminar**, virtual, 5/2021
20. **University of Iowa, Mathematical Biology Seminar**, virtual, 2/2021
21. **Northwestern University, Engineering Sciences & Applied Mathematics Colloquium**, virtual, 11/2020
22. **Brandeis University, Mathematical Biology Seminar**, virtual, 6/2020
23. **Baylor College of Medicine/Rice University, Theoretical Neuroscience Seminar**, virtual, 5/2020
24. **Colorado State University, Applied Mathematics Seminar**, Fort Collins Colorado, 2/2019
25. **Princeton Neuroscience Institute Seminar**, Princeton, New Jersey, 10/2018
26. **Institut d'Investigacions Biomèdiques August Pi i Sunyer, Computational Neuroscience Seminar**, Barcelona, Spain, 6/2018
27. **Institut de Mathématiques de Toulouse, Partial Differential Equations Seminar**, Toulouse, France, 12/2017
28. **Institut national de recherche en informatique et en automatique, MathNeuro Seminar**, Sophia Antipolis, France, 11/2017
29. **École Normale Supérieure, Neural Theory Seminar**, Paris, France, 11/2017
30. **University of Pennsylvania, Computational Neuroscience Seminar**, Philadelphia, Pennsylvania, 10/2017
31. **Colorado School of Mines, Applied Mathematics Colloquium**, Golden, Colorado, 8/2017
32. **University of Colorado School of Medicine, Physiology and Biophysics Seminar**, Aurora, Colorado, 11/2016
33. **Colorado State University, Applied Mathematics Seminar**, Fort Collins, Colorado, 9/2016
34. **University of Colorado, Applied Mathematics Colloquium**, Boulder, Colorado, 9/2016
35. **University of Arkansas, Physics Colloquium**, Fayetteville, Arkansas, 3/2016
36. **Louisiana State University School of Medicine, Cell Biology and Anatomy Colloquium**, New Orleans, Louisiana, 9/2015
37. **University of Colorado, Applied Mathematics Colloquium**, Boulder, Colorado, 11/2014
38. **University of Minnesota, Mathematical Biology Seminar**, Minneapolis, Minnesota, 11/2013
39. **University of Houston, Mathematics Colloquium**, Houston, Texas, 2/2012

40. **Hungarian Academy of Sciences, Neural Computing Seminar**, Budapest, Hungary, 11/2011
41. **Rice University, Computational and Applied Mathematics Colloquium**, Houston, Texas, 1/2011
42. **University of Nottingham, Mathematical Neuroscience Seminar**, Nottingham, United Kingdom, 11/2009
43. **Institut national de recherche en informatique et en automatique, NeuroMathComp Seminar**, Sophia Antipolis, France, 10/2009
44. **National Institutes of Health, NIDDK Laboratory of Biological Modeling Seminar**, Bethesda, Maryland, 9/2009
45. **University of Pittsburgh, Mathematical Biology Seminar**, Pittsburgh, Pennsylvania, 9/2009

Outreach, Tutorial, and Press Talks

1. **Konstanz School of Collective Behaviour**, Konstanz, Germany, 7/2026
2. **Konstanz School of Collective Behaviour**, Konstanz, Germany, 7/2024
3. **New Zealand Mathematical Research Institute Workshop on Mathematical Neuroscience**, Auckland, New Zealand, 1/2024
4. **American Physical Society March Meeting**, Denver, Colorado, 3/2020
5. **International Conference on Mathematical Neuroscience**, Copenhagen, Denmark, 6/2019
6. **Colorado School of Mines, Math Biology Summer School**, Golden, Colorado, 5/2018
7. **Houston Museum of Natural Science**, Sugar Land, Texas, 10/2014

Invited Minisymposium and Small Workshop Talks

1. **SIAM Applications of Dynamical Systems (Minisymposium): Cascade dynamics on networks**, Atlanta, Georgia, 5/2027
2. **SIAM Conference on Mathematics of Data Science (Minisymposium): Geometric and topological frameworks for spatiotemporal brain dynamics**, Salt Lake City, Utah, 11/2026
3. **SIAM Conference on Mathematics of Data Science (Minisymposium): Data science meets neuroscience**, Atlanta, Georgia, 10/2024
4. **Society for Mathematical Biology MathNeuro Mini-Conference** (Annual Meeting satellite), virtual, 6/2025
5. **SIAM Applications of Dynamical Systems (Minisymposium): Dynamics of colonies**, Denver, Colorado, 5/2025
6. **MURI Research Forum, Cognitive Fatigue, University of Michigan**, Ann Arbor, Michigan, 12/2022
7. **AMS Western Sectional Meeting (Special Session): Mathematical modeling of biological and social systems**, Salt Lake City, Utah, 10/2022
8. **AIMS Conference on Dynamical Systems (Special Session): Modeling and dynamic analysis of complex patterns in biological systems and data**, Madrid, Spain, 7/2014
9. **IMACS International Conference on Nonlinear Evolution Equations and Waves (Special Session): Mechanisms for computations in neuronal networks**, Athens, Georgia, 4/2015

10. **IMACS International Conference on Nonlinear Evolution Equations and Waves (Special Session): Dynamics of neuronal networks**, Athens, Georgia, 3/2013
11. **SIAM Conference on the Analysis of Partial Differential Equations (Minisymposium): Nonlinear dynamics of PDE in biology**, Berlin, Germany, 3/2022
12. **Joint Mathematics Meetings (Minisymposium): AMS Special Session on Stochastic Models in Studying Biological Systems**, virtual, 3/2022
13. **Society for Mathematical Biology (Minisymposium): Stochasticity and heterogeneity in network synchronization**, virtual, 7/2021
14. **SIAM Applications of Dynamical Systems (Minisymposium): Neural dynamics of sensory systems**, virtual, 5/2021
15. **Bernstein Computational Neuroscience Conference: Workshop on dynamic probabilistic inference in the brain**, virtual, 9/2020
16. **SIAM Pacific Sectional Meeting (Special Session): Theoretical neuroscience**, Seattle, Washington, 10/2019
17. **Organization for Computational Neuroscience (Workshop): How does learning reshape the dimensionality of collective network activity?**, Seattle, Washington, 7/2018
18. **SIAM Central States Sectional Conference (Minisymposium): Applied Dynamical Systems**, Fort Collins, Colorado, 9/2017
19. **SIAM Applications of Dynamical Systems (Minisymposium): Excitability, Feedback, and Collective Decision-Making Dynamics**, Snowbird, Utah, 5/2017
20. **AMS Sectional Meeting (Special Session): Mathematical neuroscience and physiology**, Pullman, Washington, 4/2017
21. **SIAM Life Sciences (Minisymposium): PRCs and Phase Models in Neuroscience**, Boston, Massachusetts, 7/2016
22. **AMS Sectional Meeting (Special Session): Nonlinear waves of differential equations**, New Brunswick, New Jersey, 11/2015
23. **SIAM Applications of Dynamical Systems (Minisymposium): Analysis of Network Dynamical Systems**, Snowbird, Utah, 5/2015
24. **IMACS International Conference on Nonlinear Evolution Equations and Waves: Special Session on Mechanisms for Computations in Neuronal Networks**, Athens, Georgia, 4/2015
25. **SIAM Life Sciences (Minisymposium): Dynamics of Multistable Perception and Decision Making**, Charlotte, North Carolina, 8/2014
26. **AIMS Conference on Dynamical Systems: Special Session on Random Dynamical Systems in the Life Sciences**, Madrid, Spain, 7/2014
27. **AIMS Conference on Dynamical Systems (Special Session): Random dynamical systems in the life sciences**, Madrid, Spain, 7/2014
28. **Conference on Frontiers in Applied and Computational Mathematics**, Newark, New Jersey, 6/2013
29. **IMACS International Conference on Nonlinear Evolution Equations and Waves: Special Session on Dynamics of Neuronal Networks**, Athens Georgia, 3/2013
30. **SIAM Life Sciences (Minisymposium): Perceptual Rivalry and Mathematical Modeling**, San Diego, California, 8/2012
31. **Canadian Applied and Industrial Mathematical Society Meeting (Minisymposium): Applied Analysis (with Dynamical Systems)**, Toronto, Ontario, Canada 6/2012

Conference Organization

- **Mathematical principles of decentralized collective decisions**, Minisymposium at SIAM Life Sciences, Cleveland, Ohio, 7/2026
- **Stochastic and dynamical models of brains and behavior**, Minisymposium at SIAM Central States Meeting, Fayetteville, Arkansas, 10/2025
- **Collective social phenomena: Dynamics and data**, Workshop at Casa Matemática Oaxaca, Oaxaca, Mexico, 6/2025
- **Dynamics Days US**, Organizing Committee, Denver, Colorado, 1/2025
- **Math + Neuroscience: Strengthening the interplay between theory and mathematics**, Semester-Long Program at ICERM, Brown University, Providence, Rhode Island, Fall 2023 (co-organizer; two workshops)
- **Dynamics of decisions and behavior in social systems**, Minisymposium at SIAM Dynamical Systems, Portland, Oregon, 5/2023
- **Dynamical principles of biological and artificial neural networks**, Workshop at Banff International Research Station, Banff, Alberta, Canada, 1/2022
- **International Conference on Mathematical Neuroscience**, Conference Chair (2017), Co-Chair (2016), Scientific/Advisory Committee (2018–2024)
- **Phase-amplitude reduction: Koopman and control**, Minisymposium at ICMN, virtual, 6/2021
- **The dynamics and limitations of working memory**, Workshop at Annual Conference on Computational Neuroscience, Barcelona, Spain, 7/2019
- **Dynamical models of individual and collective decision-making**, Minisymposium at SIAM Life Sciences, Minneapolis, Minnesota, 8/2018
- **Gulf Coast Consortium Annual Conference on Theoretical and Computational Neuroscience**, Rice University, Houston, Texas, 2/2015
- **Nonlinear and stochastic dynamics in large neuronal networks**, Minisymposium at SIAM Applications of Dynamical Systems, Snowbird, Utah, 5/2015
- **Neural mechanisms of working memory limits**, Workshop at Annual Conference on Computational Neuroscience, Paris, France, 7/2013
- **Stochasticity in large networks of the brain**, Minisymposium at SIAM Applications of Dynamical Systems, Snowbird, Utah, 5/2013
- **Spatiotemporal dynamics in networks of the brain**, Minisymposium at SIAM Life Sciences, San Diego, California, 8/2012
- **Criticality, threshold phenomena, and network dynamics**, Conference at Complex Biological Systems Group Theme Days, University of Pittsburgh, Pennsylvania, 5/2012
- **SIAM/MAA Mid-Atlantic Regional Applied Mathematics**, Student Conference at Shippensburg University, Pennsylvania, 4/2012
- **Sensorimotor processes reflected in spatiotemporal dynamics of neuronal activity**, Workshop at Cosyne, Snowbird, Utah, 2/2012
- **The role of adaptation and depression in neuronal network dynamics**, Minisymposium at SIAM Life Sciences, Pittsburgh, Pennsylvania, 7/2010
- **Cortical network dynamics**, Minisymposium at SIAM Life Sciences, Montreal, Quebec, Canada, 8/2008
- **IGERT Annual Student Workshop**, University of Utah, Salt Lake City, Utah, 5/2008

Other Trainees Supervised

13 undergraduates; 6 graduate rotations; 1 postbac

- Ishneet Chadra, **grad rotation** (CSCI), 2026
- Matthew Younce, **ugrad** APPM, 2026
- Zara Chandra, **ugrad** APPM, 2025–2026
- Emma Westerhoff, **grad rotation** ECEE, 2025
- Pedro Albuquerque Lemos, **grad rotation** (IQ Bio Program), 2025
- Rodrigo Aguayo Solis, **grad rotation** (UNAM) 2024
- James Miles, **ugrad** APPM, 2024
- Brian Tan, **ugrad** APPM/CS, 2023
- Josh Seabaugh, **grad rotation** (IQ Bio Program), 2020
- Lyanna Kessler, **grad rotation** (IQ Bio Program), 2020
- Emily Webb, **ugrad** APPM, 2019–2020
- Patrick Talley, **MSAM** APPM, 2019–2020
- Nikhil Krishnan, **ugrad** APPM, 2017–2018
- Elliott Saslow, **ugrad** MCDS, 2017
- Matthew Hansen, **ugrad** APPM, 2016–2017
- Jacob Parelman, **postbac** Psych, 2017
- Courtney Van Den Elzen, **grad rotation** (IQ Bio), 2017
- Nicholas Troutman, **ugrad** (U Houston), 2015
- Zachary McCleney, **ugrad** (U Houston), 2015
- Sam Carroll, **ugrad** (U Houston), 2013
- Stephanie Willoughby, **ugrad** (Ohio State), 2013
- Shawn Gu, **ugrad** (Ohio State), 2013
- Kate Nguyen, **ugrad** (U Houston), 2014 **(2014 Goldwater Scholar)**
- Mahjub Hammond, **ugrad** (U Pittsburgh), 2012
- Suren Jayasuriya, **ugrad** (U Pittsburgh), 2010–12

Dissertation Committees

- Emily Ekstrum, Computational Biosciences (CU School of Medicine), 2030
- Luc Brancheau, Math (NJIT), 2029
- Emma Keppler, NEUR (CU Boulder), 2029
- Poom Kritpracha, APPM (CU Boulder), 2028
- Jonathan Homan, APPM (CU Boulder), 2028
- Sean Svilha, APPM (CU Boulder), 2028
- Erin Obermayer, APPM (CU Boulder), 2027
- Nicholas Garcia, Computational Biosciences (CU School of Medicine), 2027
- Madi Yerlanov, APPM (CU Boulder), 2026
- Christina Wang, APPM (CU Boulder), 2026
- Siddharth Paliwal, Neuroscience (SUNY Stonybrook), 2025
- Yassin Bahid, APPM (CU Boulder), 2025
- Colin Korbisch, MCEN (CU Boulder), 2025
- Corbit Sampson, APPM (CU Boulder), 2025
- Sabina Adhikari, APPM (CU Boulder), 2024
- Rachel Robey, APPM (CU Boulder), 2024
- Jamie Voros, ASEN (CU Boulder), 2023
- Amanda Hampton, APPM (CU Boulder), 2023
- Megan Stickler, Mathematics (U Houston), 2022
- Nicholas Landry, APPM (CU Boulder), 2022
- Samuel Ryskamp, APPM (CU Boulder), 2022
- Lyndsey Wong, APPM (CU Boulder), 2022
- Erin Ellefsen, APPM (CU Boulder), 2022
- Sabina Altus, APPM (CU Boulder), 2021
- Shelly Jones, Neuroscience (CU School of Medicine), 2020
- Harry Dudley, APPM (CU Boulder), 2020
- Jaqueline Wentz, APPM (CU Boulder), 2020
- Elijah Christensen, Neuroscience (CU School of Medicine), 2020
- Joshua Aurand, APPM (CU Boulder), 2020
- Callie Federer, Computational Biosciences (CU School of Medicine), 2019
- Sama Shretha, APPM (CU Boulder) 2019
- Jay Stotsky, APPM (CU Boulder), 2018
- John Nardini, APPM (CU Boulder), 2018
- Wei-Ting Li, Biology (UH), 2017

- Inomzhon Mirzaev, APPM (CU Boulder), 2017
- Changan Liu, Mathematics (UH), 2017
- Jose Manuel Lopez, Mathematics (UH), 2014

Masters Thesis Committees

- Rachel Rise, ASEN (CU Boulder), 2021
- David Stearns, APPM (CU Boulder), 2021
- Jamie Voros, ASEN (CU Boulder), 2020
- Kadambari Suri, ASEN (CU Boulder), 2019

Undergraduate Honors Thesis Committees

- Annika Beighey, Neuroscience (CU Boulder), 2027
- Jun Means, Neuroscience (CU Boulder), 2026

Reviewing and Editing

- **Editor:** *SIAM Journal on Applied Dynamical Systems (2025–)*; *Mathematical Medicine & Biology (2024–)*; *Mathematical Neuroscience & Applications (2021–)*; *SIAM Dynamical Systems Web Magazine (2022–2023)*; *Journal of Mathematical Neuroscience (2017–2019)*
- **Proposal Review Panelist:** *NSF (CAREER; Mathematical Biology; MathBioSys; Dynamics, Control, and System Diagnostics; Robust Intelligence; Physics of Living Systems; MSPRF; CRCNS); NIH (BRAIN Initiative: Theories, Models and Methods for Analysis of Complex Data from the Brain); Agence Nationale de la Recherche (France); Wellcome Trust Fellowships (UK)*
- **Book Reviewer:** *SIAM* and *Taylor & Francis*
- **Conference Abstract Reviewer:** *Cosyne (2014, 2017–)* and *International Conference on Mathematical Neuroscience (2016–)*
- **Referee for hundreds of submitted manuscripts.**
  Multidisciplinary: *eLife; iScience; Nature Communications; Nature Reviews Neuroscience; PLOS One; Proceedings of the National Academy of Sciences; Scientific Reports*
  Neuroscience and quantitative biology: *Biological Cybernetics; Biophysical Journal; Frontiers in Computational Neuroscience; Frontiers in Systems Neuroscience; Journal of Computational Neuroscience; Journal of Mathematical Neuroscience; Journal of Neurophysiology; Journal of Neuroscience; Neural Computation; Neural Networks; Neurocomputing; PLOS Computational Biology*
  Physics: *New Journal of Physics; Physica D; Physical Review E; Physical Review Letters; Physical Review Research; Physical Review X; PRX Life*
  Applied mathematics and dynamical systems: *Chaos; Discrete and Continuous Dynamical Systems Series B; European Journal of Applied Mathematics; Journal of Mathematical Biology; Nonlinearity; SIAM Journal on Applied Dynamical Systems; SIAM Journal on Applied Mathematics; SIAM Journal on Mathematical Analysis; SIAM Review*

Other Committee Work

- **Faculty Search Committee**, Chair, CU Boulder, Department of Applied Mathematics, 2026–2027
- **Colloquium Chair**, CU Boulder, Department of Applied Mathematics, 2017–2018, 2026–2027
- **Space Stewardship Task Force**, University of Colorado Boulder, 2026
- **BioDataSci Academic Advisory Committee**, CU Boulder, 2025–

- **IQ Biology PhD Application Review Committee**, CU Boulder, 2024–2026
- **Computational Neuroscience Cluster, Hiring Committee**, CU School of Medicine, Department of Physiology & Biophysics, 2023–
- **Affiliate Faculty Committee**, CU Boulder, Department of Applied Mathematics, 2023–2026
- **College of Arts & Sciences Faculty Senate, Budget Committee**, CU Boulder, 2020–2026
- **Graduate Partial Differential Equations Exam Committee**, CU Boulder: 9 times
- **Chair's Advisory Committee**, CU Boulder, Department of Applied Mathematics, 2021–2026
- **PUEC Promotion & Tenure Committee**, CU Boulder, Department of Applied Mathematics, 2021, 2023, 2026
- **Tech Frontiers Program**, CU Boulder, Department of Computer Science, 2021–2023
- **SIAM Dynamical Systems Group**, Secretary, 2022–2023
- **IQ Biology Academic Advising Committee**, CU Boulder, BioFrontiers Institute, 2018–2022
- **Joint APPM/CSCI Hiring Committee**, CU Boulder, Department of Applied Mathematics, 2021
- **Graduate Committee**, CU Boulder, Department of Applied Mathematics, 2017–2021
- **PUEC Reappointment Committee**, CU Boulder, Department of Applied Mathematics, 2020
- **APPM 30th Anniversary Celebration Committee**, CU Boulder, 2019
- **CAES-APPM Partnership Committee**, CU Boulder, 2017–2018
- **Awards Committee**, CU Boulder, Department of Applied Mathematics, 2016–2017
- **Graduate Committee**, UH, Department of Mathematics, 2014–2015
- **Gulf Coast Consortium for Theoretical and Computational Neuroscience**, UH/Rice University/Texas Medical Center, 2012–2016
- **Colloquium Committee**, UH, Department of Mathematics, 2012-2016
- **NETWORKS Seminar Committee**, UH, 2012–2016

Outreach

- **CU Boulder Leadership, Education, Advancement, & Promotion**, panelist: 2025, 2026
- **CU Boulder Graduate Inclusive Mentorship Community**, participant, 2024-2025
- **SIAM Dynamical Systems Group Mentoring Program**, panelist, 2021
- **National Alliance for Doctoral Studies in the Mathematical Sciences**, mentor, 2014–
- **Association for Women in Math**, U Utah, alumnus mentor, 2016–2017
- **Summer Undergraduate Research Fellowship**, UH, professional development panelist, 2015
- **SIAM/AMS Student Chapter**, UH, professional development panelist, 2013–2016
- **Cougar and Houston Area Mathematics Program (CHAMP)**, UH, facilitating high school mathematics outreach program, 2013–2016

Consulting Activities

2020– Data Science Instructor and Consultant, **Data Society**, Washington DC
2020 Consultant Scientist, **Allen Institute: Mindscope Program**, Seattle WA
2018 Scientific & Technical Consultant, **FullContact**, Denver CO